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RESEARCH ARTICLE

Optimal Sizing and Improved Low-Pass Filter Energy Management for Hybrid Energy Storage Electric Vehicles

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ABSTRACT The global promotion of electric vehicle (EV) usage is a response to the challenges of energy crises and climate change. However, significant drawbacks remain, particularly related to the high costs and limitations of onboard energy storage. To mitigate these issues, the concept of Hybrid Energy Storage Systems (HESS), which integrate batteries with complementary energy storage solutions, has gained attention. This study focuses on the optimal sizing and energy management strategy of HESS, which are critical for enhancing the performance and viability of these systems in EVs. An Improved Low-Pass Filter (ILPF) is introduced to address limitations of conventional low-pass filters (LPF), including phase shift effects and state of charge limitations while optimizing the cut-off frequency and ensuring adaptability to varying driving conditions. Particle Swarm Optimization (PSO) is employed to achieve optimal sizing of the HESS and to develop an efficient energy management strategy (EMS). The performance of the ILPF is validated against a Fuzzy Logic Controller (FLC) and a conventional-LPF across various driving cycles. Simulation results demonstrate that the proposed ILPF significantly enhances performance, achieving up to an 8.345% improvement in city driving conditions compared to battery-only electric vehicles.

INDEX TERMS Electric vehicles, energy management, hybrid energy storage, optimization, sizing.

NOMENCLATURE

BMS	Battery Management System.
CoA	Center of Area.
CPS	Cycle per Second.
CS	Component Sizing.
DoD	Depth of Discharge.
DP	Dynamic Programming.
EMS	Energy Management Strategy.
ESS	Energy Storage Systems.
EV	Electric Vehicle.
FLC	Fuzzy Logic Controller.
GA	Genetic Algorithm.
GBM	Generic Battery Model.
GWO	Grey Wolf Optimization.
HESS	Hybrid Energy Storage Systems.
HPF	High-Pass Filter.
HWFET	Highway Fuel Economy Test Cycle.
ILPF	Improved Low-Pass Filter.
LPF	Low-Pass Filter.
OLPF	Optimal-LPF.
OLPF-PSR	OLPF with PSR.
PMSM	Permanent Magnet Synchronous Motor.
PSO	Particle Swarm Optimization.
PSR	Phase Shift Reducer.
RF	Random Forest.
SA	Simulated Annealing.
SC	Supercapacitor.

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SDP	Stochastic Dynamic Programming.
SF	Scaling Factor.
SoC	State of Charge.
UDDS	Urban Dynamometer Driving Schedule.
WLTC	Worldwide Harmonized Light Vehicles Test Cycle.
WLTP	Worldwide Harmonized Light Vehicle Test Procedure.

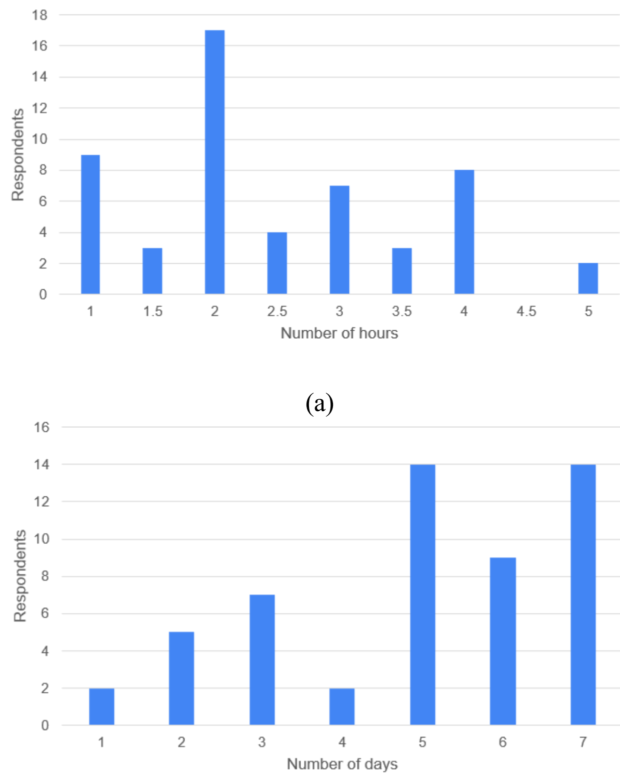
I. INTRODUCTION

The global automotive industry is experiencing a transformative shift towards sustainable mobility solutions in response to environmental concerns and escalating energy demands. Central to this transition are EVs, which offer lower emissions, improved energy efficiency, and reduced dependence on traditional internal combustion engines [1]. Many countries worldwide have promoted the growth of electric vehicle usage to address the challenges posed by the energy crisis and global warming [2]. Nevertheless, there are several drawbacks to EVs, including the overall cost of the car and, most importantly, the limitations of onboard energy storage [3]. Therefore, the concept of HESS is proposed by researchers which combines batteries with other energy storage systems to solve battery issues. Given that batteries are among the most expensive components of EVs, prolonging their lifespan can help them become more cost-effective over the long run. The HESS can improve overall system performance, reduce expenses, and boost efficiency [4].

A preliminary survey conducted by authors in Indonesia revealed that vehicles are primarily used for approximately two hours a day, with utilization occurring five to seven days a week. The survey results of 53 respondents are shown in Fig. 1. This data suggests a conservative worst-case scenario for battery usage of five hours daily and seven days a week, translating to an annual utilization of 365 days. Understanding these usage patterns highlights the importance of optimizing the design of HESS to enhance battery lifespan and overall vehicle efficiency.

Optimal sizing and efficient energy management are critical factors influencing the performance and viability of HESS EVs. HESS, consisting of batteries, supercapacitors (SC), and other energy storage elements, plays a pivotal role in managing energy flows, storing regenerative braking energy, and providing power during acceleration phases. Effective optimization of HESS can enhance vehicle performance, extend battery life, and minimize energy consumption.

Despite advancements in HESS, notable gaps remain in the literature. Many existing studies focus primarily on component sizing and energy management strategies but often treat these processes separately, leading to suboptimal performance and increased costs [5]. The sizing process of HESS is also dependent on the power allocation method [6], [7]. While various optimization methods have been employed, including iterative techniques and Genetic Algorithms (GA), the rationale for selecting PSO in this study is based on its



(a)

(b)

FIGURE 1. Car utilization survey results: (a) hours per day, (b) days per week.

advantages in efficiency, resilience, convergence, and flexibility, particularly in addressing the complexities of HESS under severe constraints [8]. Additionally, existing works primarily address the sizing of HESS without adequately considering the sizing of the DC-DC converter, which is essential for overall system design.

Within the HESS, the EMS is crucial [9]. It is employed to disperse the electricity demand among several Energy Storage Systems (ESS). The EMS approach is divided into three categories: rule-based, optimization-based, and learning-based [5]. The rule-based approach can be used in real-time since it is simple and requires little computing time, as stated in [10] and [11]. The Honda Insight and Toyota Prius both employ this tactic [12]. Since it is the most popular and straightforward method based on [13], the LPF technique, a part of rule-based EMS, is utilized in this work. However, challenges associated with LPF energy management exist, including inadequate control over the state of charge (SoC), difficulties in finding the appropriate cut-off frequency, phase shift issues that can cause unnecessary energy exchange between energy storage systems, and the need for adaptability to changing conditions [14], [15].

There are numerous methods available for achieving optimal sizing of HESS and effective LPF EMS. The optimal LPF EMS means that the cut-off frequency is derived from the optimization process [14]. Some of the optimization methods used are iterative [9] and dynamic programming (DP) [16] [17]. Scholars have proposed some component sizing (CS) methods. Authors in [18] classify the sizing method as traditional and optimization-based. Experience and equivalent calculation are part of the traditional one. The methods that belong to optimization are iterative, GA, and PSO. Reference [19] combines Grey Wolf Optimization (GWO) and DP for CS and EMS respectively. Further, they use Random Forest (RF) to learn the rules from DP to be implemented in real-time. Authors in [20] do the same concept and replace GWO with Simulated Annealing (SA) and RF with Stochastic DP (SDP). GA and FLC as CS and EMS, respectively, are mostly used in [21], [22], and [23].

Most existing studies focus solely on the sizing of HESS without considering the DC-DC converter size. Therefore, this study proposes a comprehensive approach that integrates optimal sizing of both HESS and the DC-DC converter with an effective LPF EMS. By deriving the sizing and optimal LPF EMS simultaneously through the PSO process, this work corroborates and builds upon existing research by addressing both HESS sizing and the critical role of the DC-DC converter in the system's performance. In terms of efficiency, resilience, stagnation, convergence, and sizing flexibility under severe constraints, PSO outperforms both DP and GA [8]. Moreover, its performance in HESS sizing and management has already been demonstrated by prior studies [24], [25].

The outcomes of these efforts demonstrate significant improvements in the performance and sustainability of HESS EVs. By optimizing the sizing and energy management of HESS EVs, this study contributes by:

1) Proposing a simple sizing method for HESS based on the total traveling distance of electric vehicles while minimizing component costs.
2) Addressing challenges in the LPF EMS. This includes finding the optimal cut-off frequency, implementing effective state of charge control, reducing phase shift effects, and improving adaptability to varying driving conditions.

The remainder of the article is organized as follows: Section II reviews the electric vehicle model. Section III discusses the proposed optimal sizing and energy management strategy. Section IV presents the testing findings and analysis. Finally, Section V concludes the article.

II. HESS EV MODEL

In this study, a city car with the specifications in Table 1 is used which was adopted from [26] and modified. The HESS consisting of a lithium battery and SC is used as the hybrid power source. The battery is a primary source whereas the SC is the secondary source. Based on the energy source classification, the battery has high energy, and SC is in the high-power category. The high energy can give long energy but with a small amount of power, hence it is suitable for long-distance travel. On the opposite, SC as high power can charge and discharge with high current but for a moment hence it is suitable for acceleration and deceleration.

Since the primary energy source is the battery, its specification is calculated based on the power required by the EV system. The use of SC is to prolong the battery life not for increasing the vehicle traveling distance. The price of SC is higher than that of a lithium battery; therefore, low-energy SC is used, and it is only enough to cut peak power and has only a small effect on the vehicle's traveling distance. The battery and SC specification used is listed in Table 2. The SC pack is designed based on the combination of the SC module with the optimal size which is obtained from the optimization process. The SC optimal sizing will be discussed in the next section.

TABLE 1. HESS EV parameters.

Parameters	Value	Unit
Vehicle mass without ESS	500	kg
Driver mass	75	kg
Windward area (Af)	2.04	m^2
Gravitational acceleration (g)	9.81	m/s^2
Air density (ρ_{air})	1.25	kg/m^3
Rolling resistance coefficient (c_{roll})	0.0112	-
Air resistance coefficient (Cd)	0.25	-
Wheel radius (r_w)	0.252	m
Transmission ratio (G)	6.5	-
Traction motor power (P_{m_max})	30	kW
Traction motor torque (T_{m_max})	110	Nm
Traction motor voltage	144	V
Inverter efficiency (η_{inv})	0.95	-
DC-DC converter efficiency	0.95	-
Auxiliary load (P_{aux})	500	W

TABLE 2. Battery and SC module specification.

Battery Module [27]	
Nominal voltage	12.8 V
Voltage range	8-15.6 V
Nominal capacity	7.5 Ah
Weight	1.1 kg
Internal resistance	0.06 Ω
Cycle life (80% DOD)	3000
c-rate (discharge/ charge)	1c/ 0.5c
SC Module [28]	
Capacitance	58 F
Voltage	16 V
Maximum continuous current	23 A
Cycle life	500,000

The required battery energy (E_B) is calculated as

$$E_B = \left(\frac{E_{avg} * D}{\eta_t}\right) * (2 - 80\%\text{DoD}) \tag{1}$$

where E_{avg} (Wh/km) is the average energy consumption of the EV, in this study, measured in the Worldwide Harmonized Light Vehicles Test Cycle (WLTC) drive cycle, while D and η_t are the traveling distance requirement and the total system efficiency, respectively. Since only 80% of the battery's depth of discharge (DoD) is used, then terms (2 – 20% DoD) are added. With the $Eavg$ of 68.63 Wh/km, D value of 200 km, and the η_t of 0.9, the E_B is 18.30 kWh.

The DC-bus voltage is chosen as the same as the nominal voltage of the traction motor. Therefore, the series of battery modules is $Ns = 144/12.8 = 11.25$ rounded to 12 modules. Hence the voltage of the battery pack (V_{B-pack}) is 153.6 V. The number of batteries in parallels (N_P) is calculated as

$$N_P = (E_B/V_{B-pack})/C_B \tag{2}$$

where C_B is the battery's nominal capacity. It gives the N_P value of 16 modules. Finally, the battery pack has the specifications of 153.6 V, 120 Ah, and 18.43 kWh. Battery mass is calculated as

$$M_{B-pack} = Ns * Np * M_B \tag{3}$$

where M_{B-pack} and M_B are the mass of the battery pack and battery module, respectively. The resulting M_{B-pack} is 211.20 kg. With the addition of a Battery Management System (BMS) and casing, the total battery mass is assumed to be 250 kg.

The HESS EV control system structure used is shown in Fig. 2. Inside the blue box is for HESS whereas the other is the speed control of the Permanent Magnet Synchronous Motor (PMSM) including the vehicle load named Glider. In the HESS there is only one DC-DC bidirectional converter on the SC side known as semi-active topology. This topology is the tradeoff between the price and control ability. The passive topology is the cheapest one, but it lacks power control. On the other hand, the active topology which uses a DC-DC converter in each ESS is the most expensive, even though it has the best control ability for power sharing. The structure of the semi-active HESS is illustrated in Fig. 3.

The model utilizes MATLAB software's Generic Battery Model (GBM) and uses a lithium-ion (Li-ion) battery as its power source. The SC model uses the non-linear Stern-Tofel model. For more details about the battery and SC model used see [29]. The DC-DC converter is modeled as constant efficiency. To ensure that the battery operates within the SoC limits, the responsibility typically lies with the BMS in real-world electric vehicles. In this simulation,

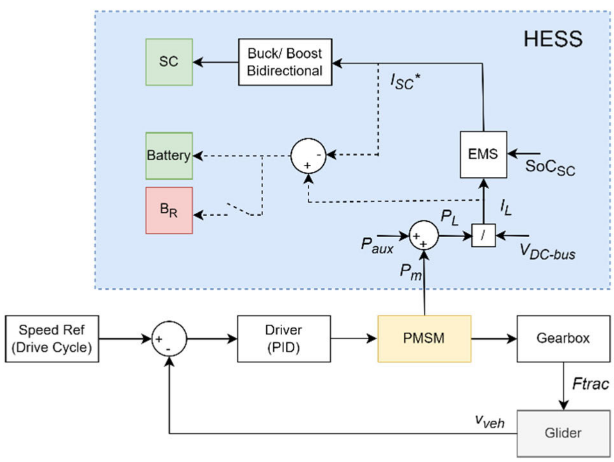


FIGURE 2. Proposed HESS EV control system structure.

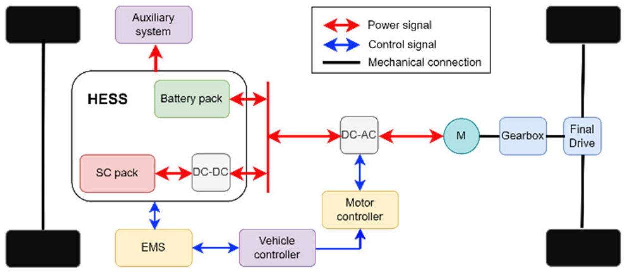


FIGURE 3. Semi-active HESS EV configuration used, adopted from [26].

however, a BMS model is not utilized. Instead, a protocol has been established that halts the simulation when the battery SoC falls below 20%. For the upper limit, a braking resistor (B_R) has been implemented, which activates if the reference current for the battery exceeds safe limits or when the battery is fully charged in the braking mode. The power limitation is also added to limit the power requested by the vehicle load under the battery's maximum power. This mechanism ensures that the battery operates under the maximum current as specified. Hence, the BMS in real vehicles does not activate the protection of over-voltage or over-current.

The EMS distributes the load power (P_L) which consists of motor load (P_m) and auxiliary load (P_{aux}) in the form of load current (I_L) after dividing the load power with the DC-bus voltage (V_{DC-bus}). The output of the EMS system is the current reference for the SC. The rest of the load current is for the battery, the dashed line (Fig. 2) means that the battery references current is uncontrol. The traction motor uses a PMSM. For simplicity, the PMSM with vector control is modeled as a transfer function as [30]:

$$\mathrm{TF_{PMSM}} = \frac{\omega}{(T_e - T_L)} = \frac{1}{Js + B} \tag{4}$$

where ω, T_e, T_L, J, and B are angular speed, electrical torque, load torque, rotor inertia, and viscous friction coefficient, respectively. At the same time, the inverter is modeled as constant efficiency. The Glider as the vehicle load is modeled using Newton's motion law as

$$\frac{dv_{veh}}{dt} = (F_{trac} - F_{roll} - F_{aero} - F_{grade}) \div M_{veh} \tag{5}$$

$$F_{roll} = c_{roll} M_{veh} g cos(\delta) \tag{6}$$

$$F_{aero} = \frac{1}{2}\rho_{air} A_f C_d v_{veh}^2 \tag{7}$$

$$F_{grade} = M_{veh} g \sin(\delta) \tag{8}$$

$$P_m = (T_e \omega)/\eta_{inv} \tag{9}$$

where the traction force (F_{trac}) is the total force required to move the vehicle. There are three motion resistances accommodated which are rolling resistance (F_{roll}), aerodynamic resistance (F_{aero}), and gradient resistance (F_{grad}). The vehicle speed and total mass are denoted as v_{veh} and

M_{veh}, respectively. The parameter's value is listed in Table 1. The motor request power is calculated using (9).

The structure of the LPF EMS employed in this study is depicted in Fig. 4. This system utilizes a semi-active configuration with a single DC-DC converter on the SC side, generating the current reference for the SC as its output. The input to the EMS is the load current (I_L), while the LPF provides a low-frequency output for the battery. The difference between the load current and the LPF output (I_{SC0}) is utilized, reflecting the DC-DC converter's position on the SC side. The LPF was selected over a High-Pass Filter (HPF) due to the simplicity of the tuning process for determining the cut-off frequency, which begins at 0 Hz. Prior to being directed to the DC-DC converter as the SC current reference (I_{SC}*), the output is processed through a rule-based mechanism to implement the SoC limiter. Additionally, the proposed method incorporates the Phase Shift Reducer (PSR) algorithm.

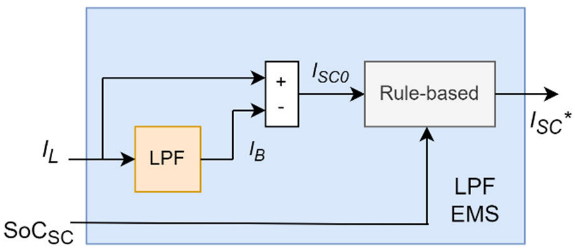


FIGURE 4. LPF EMS structure.

The FLC EMS according to [31] is commonly used as EMS for HESS EV. Therefore, it is used for comparison. Fig 5 shows the FLC EMS structure which employs Mamdani-type fuzzy. The load current (I_L) and SoC_{SC} are the inputs of the FLC, and the output is the scaling factor (SF) of the SC. The current reference for SC is then obtained by multiplying the SF by the load current. To double protect the SC SoC within the allowed range, the SoC limiter, in the form of a rule-based, was also introduced. The final SC's current reference (I_{SC}*) is then sent to the DC-DC converter.

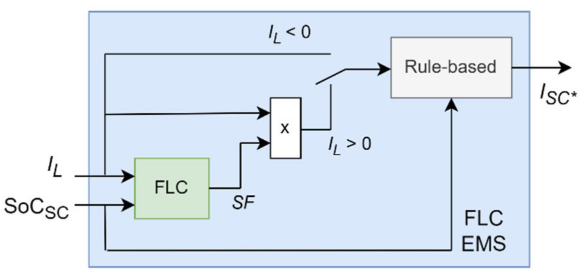


FIGURE 5. FLC EMS structure.

The fuzzy rules that are selected manually are displayed in Table 3. Small, Medium, and Big are the membership functions for S, M, and B, respectively. The criteria were developed using the assumption that SC power allocation is linear with load current and SoC_{SC}. For instance, the SC contributes more current when the higher the load current. There are nine rules in all. Processing time will rise with the number of rules. The center of area (CoA) defuzzification technique is employed. The membership function of both input and output is triangular.

TABLE 3. FLC EMS rules.

I_L | SoC SC	S	M	B
S	S	M	M
M	M	M	B
B	M	B	B

In this case, the FLC EMS has so many properties such as nine rules, and a minimum of nine parameters for input and output membership (three for each input/ output) for the optimization process. In comparison, the LPF only uses a single parameter in the optimization process. Besides the LPF being simple, it can solve the high-frequency load current is the main reason it is used.

III. SIZING AND ENERGY MANAGEMENT OF HESS

A. HESS SIZING

The optimal sizing goal in this study is to get the optimal size of the DC-DC converter and SC which can give the highest total traveling distance with minimum cost. Some researchers use mass and volume constraints. On the other hand, this constraint is not used in this study. Although the mass of the SC and its converter also affect the performance, since in this study we assume to design a new vehicle, the volume and mass constraint is not used if it gives good performance. Fig. 6 illustrates the effect of vehicle mass on EV performance in terms of estimated battery lifetime and autonomy for battery-only EVs. This graph is taken from the EV model when using a battery-only. When the vehicle mass increases, both the estimated battery lifetime and autonomy will be decreased. This data shows that the increase in vehicle mass of 75 kg (one passenger) will decrease battery lifetime and autonomy by an average of 8.715% and 6.602%, respectively. Therefore, the vehicle mass due to the use of HESS needs to be considered.

Various methodologies have been proposed for the optimal sizing of HESS that utilize both batteries and supercapacitors. For instance, Sadoun et al. propose a sizing approach based on power thresholds, which limits the power supplied by the battery while assigning excess demand to the supercapacitor [32]. In this framework, the rule-based EMS is used. The supercapacitor is engaged for loads exceeding 40 kW and below -10 kW as illustrated in Fig. 7. This strategy is similarly employed by Rizoug et al., who emphasize the need for effective EMS to optimize battery life and overall system efficiency [33]. Additionally, the authors in [34] determine the size of the supercapacitor specifically for capturing energy from regenerative braking, maximizing its role in energy recovery. In this case, the FLC EMS is used.

In contrast, this study emphasizes the economic implications of supercapacitor usage, as their costs are typically

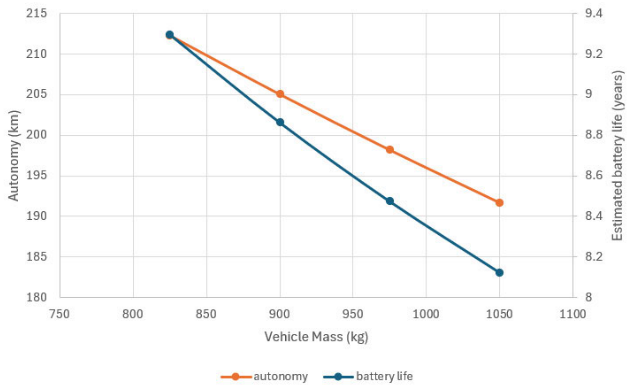


FIGURE 6. Effect of vehicle mass on its performance.

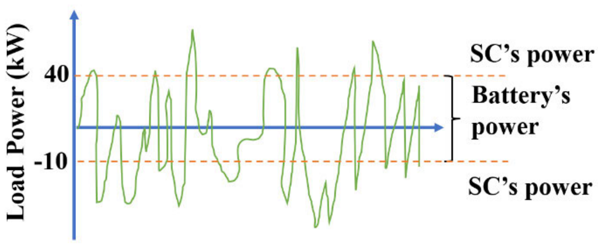


FIGURE 7. HESS sizing based on power threshold.

higher than those of batteries for equivalent energy storage. Therefore, this research prioritizes the supercapacitor's contribution to power delivery, allowing for the use of smaller, more cost-effective supercapacitors. Moreover, the integration of a DC-DC converter is considered essential in forming a complete HESS for electric vehicles. By employing PSO, this study presents a comprehensive sizing method that not only optimizes the HESS but also accounts for the critical role of the DC-DC converter, ensuring enhanced efficiency and performance.

The DC-DC converter used has a voltage input range of 90-180 V and a maximum current of ± 100 A. At the same time, the output voltage range is 120-180 V. The maximum SC's current is calculated based on the voltage at 50% SoC. The SC's voltage (Vsc) at 50% SoC is measured when the SC at 50% SoC, based on [35]:

$$\mathrm{SoC}_{sc} = V_{sc}/V_{sc-max}. \tag{10}$$

Based on the SC specification in Table 2 and the DC-DC converter specification, it requires 11 SC modules in series to get 176 V. The higher voltage is chosen to get a lower current. At this condition, the Vsc at 50% SoC is 88 V; hence the maximum DC-DC converter power is 8800 W (rounded to 9000 W) to meet the current limit. The number of parallel SC will be determined in the optimal sizing process.

B. IMPROVED LOW-PASS FILTER EMS

In the HESS EV, EMS is important to distribute the load power between the ESS uses. There are many EMS methods proposed; however, not all of them can be implemented due to computational burden. LPF EMS is simple and implementable [36]. While according to [4], it is simple, robust, and gives good results on dynamics and cycle reduction. The working principle of LPF as EMS is to decouple the low frequency from the load current and send it to the battery [37]. At the same time, the rest with higher frequency is sent to the SC. Based on the working principle, it cannot work alone as EMS. The SoC Limiter must be added to keep the ESS working at the safe SoC limit [38]. Besides that, the challenges of the application of LPF EMS are resumed in Fig. 8.

Most studies about LPF EMS in HESS EVs focus on finding the optimal cut-off frequency such as in [9] and [16]. While the other adds the adaptive capability for load changes as in [39] and [40]. As far as the authors know from the literature study, there are no references to solve the phase shift issue of LPF EMS in-vehicle applications. Since this effect contributes to significant losses, this study will accommodate to reduce this effect.

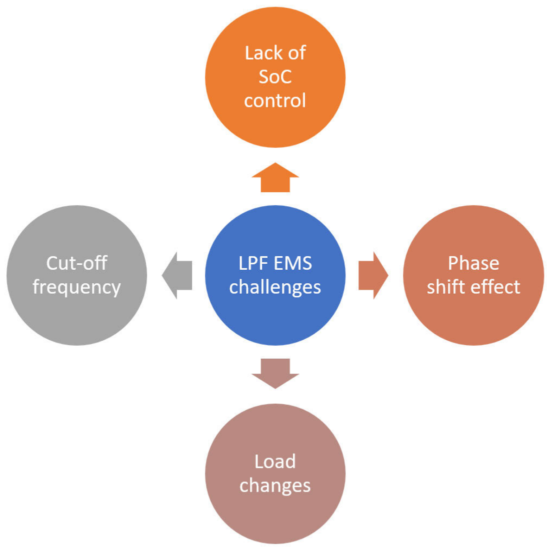


FIGURE 8. LPF EMS challenges.

One of the effects of filtering is the phase shift. In the application of LPF as EMS, the phase shift will cause unnecessary power exchange of the ESS in the HESS, illustrated in Fig. 9. In powering both the battery and SC give the current to the EV system through the DC-bus, SC has a faster response than the battery. When braking occurs, since the battery has a slow response, when the load current and SC current are negative, the battery current is still positive. This condition makes the battery give the current to the SC. This happens naturally, uncontrol, since the higher voltage will charge the lower one. This makes unnecessary energy exchange between the battery and SC happen frequently which will cause losses in the DC-DC converter.

Authors in [15] propose a rule-based method to address the challenges of LPF EMS in the application of HESS in micro-grids. The proposed method includes various operational

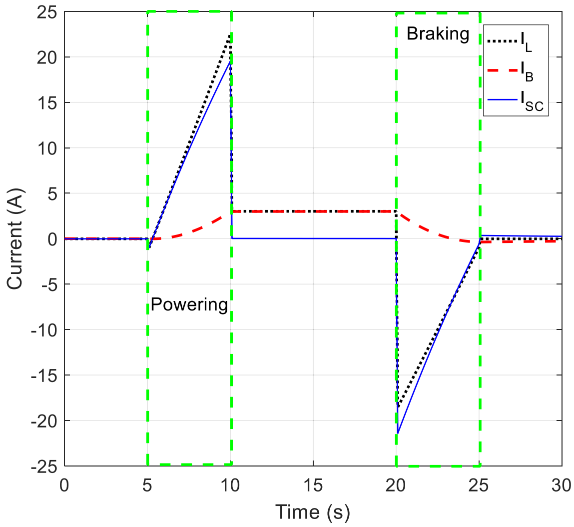


FIGURE 9. The current profile of HESS for conventional LPF EMS.

modes: Mode-I serves as the default mode where both the battery and SC provide or receive power. Mode-II applies when the battery power exceeds the HESS power, allowing the battery to charge the SC. Conversely, Mode-III is activated when the SC has higher power than the HESS, enabling the SC to charge the battery. They show the effectiveness of the rule-based in reducing the phase-shift effect of the LPF EMS. Hence, the rule-based method is adopted.

The simple rule-based method is proposed based on the same objective to reduce phase shift effects and add a SoC limiter. Its pseudocode is shown in Table 4, where I_{SC0}, *Iconv*, I_L, $I_{SC}{}^*$, and SoC_{SC} are the current reference for SC before rule-based, the current limit of DC-DC converter, load current, the current reference for SC after rule-based, and SoC SC, respectively. Starting by limiting the I_{SC0} under the converter power limit. If it is higher or lower than the converter current limit, then the maximum converter current value will be used. If the I_{SC0} is already under the converter current limit, it is divided into powering and braking conditions. Both in powering and braking, the I_{SC0} must not exceed the load current (higher for powering and lower for braking). Then SoC limiter is applied. The SoC_{SC} is limited to 55–95% to prevent SC from being overcharged and over-discharged. According to [39] the SC charge cannot be allowed to go below 50% of the maximum voltage. Therefore, the limit value is chosen at lower than 100% and higher than 50% to provide an additional safety margin. Maintaining the supercapacitor charge above this level is crucial for several reasons: it helps stabilize the voltage, enhances efficiency, extends the device's lifespan, and improves overall safety.

Next, the optimal cut-off frequency and adaptability issue is solved by PSO. It is not purely adaptive, since the cut-off frequency is not changing every time. However, it adopts the gain scheduling concept where in a specific condition it uses specific gain, in this case, the gain is the cut-off frequency. The most used adaptive criteria for LPF cut-off frequency in HESS battery-SC are load current and SC SoC [14].

TABLE 4. Pseudocode of PSR and SoC control.

```
If Isc0 > (-Iconv) and Isc0 < Iconv
   If IL > 0
      If Isc0 < IL and Isc0 > 0
         If SoCsc > 55
            Isc* = Isc0
         Else
            Isc* = 0
         End
      Elseif Isc0 > IL
         Isc* = IL
      Else
         Isc* = 0
      End
   Else
      If Isc0 > IL and Isc0 < 0
         If SoCsc < 95
            Isc* = Isc0
         Else
            Isc* = 0
         End
      Elseif Isc0 < IL
         Isc* = IL
      Else
         Isc* = 0
      End
   End
Elseif Isc0 > Iconv
   Isc* = Iconv
Elseif Isc0 < (-Iconv)
   Isc* = - Iconv
End
```

According to [31], the gradient changes, the vehicle's mass changes, and driver behavior can affect the load current. Therefore, load current is used as the adaptation parameter. The powering-braking base is a good choice for an adaptive FLC EMS based on [31]. Therefore, this study uses load current parameters which differentiate between powering and braking conditions. The threshold is set at $I_L = 0$ A, and when $I_L < 0$ then fc_1 is used, and if $I_L > 0$ then fc_2 is activated. Since this is LPF, the initial fc is chosen near 0 Hz. The maximum fc value is set to be 0.01 Hz which is the cut-off frequency from the Ragone plot method [38].

The objective function of the PSO is

$$J_{PSO} = -D_T + P_{HESS} \tag{11}$$

which is the trade-off between total traveling distance (D_T) and HESS cost (SC and DC-DC converter price - P_{HESS}). Its goal is to maximize the total traveling distance and minimize the HESS cost, hence, the minus sign is added to the total traveling distance and PSO is used as a minimization algorithm.

The total traveling distance is the total distance that can be covered by the EV as long as the battery service life. Therefore, it is calculated by the multiplication of vehicle autonomy (A_{veh}) and the estimated battery lifetime (B_{LT}) as

$$D_T = A_{veh}\left(\frac{km}{charge}\right) * 365\left(\frac{charges}{year}\right) * B_{LT}(years) \tag{12}$$

$$A_{veh} = \frac{(E_{HESS}/D)}{80\% E_B} \tag{13}$$

It assumed that the EV is charged once every day (365 charges/ year). The vehicle autonomy is calculated based on the energy consumed by HESS EV (E_{HESS}), in this case, the sum of battery and SC energy, which is divided by the traveled distance (D) to get efficiency (kWh/km) then it divided by 80% of battery energy (E_B). The value of 80% is used since the DoD of the battery is 80%.

The battery lifetime (B_{LT}) is calculated based on the self-discharge aging and cycle aging as

$$B_{LT} = (B_{shelf-discharge\ aging} + B_{cycle\ aging}) * 365(years) \tag{14}$$

$$B_{shelf-discharge\ aging} = B_{SDR}/30(days) \tag{15}$$

The lithium battery self-discharge rate(B_{SDR}) is 2% according to [41]. The self-discharge rate is used because the worst-case scenario of aging when the car is parked is accommodated as in (15). The battery cycle aging is calculated using the equivalent cycle number as

$$B_{cycle\ aging} = \text{CPS} * 3600 * 5\,(days) \tag{16}$$

$$C = \frac{1}{E}\int_0^t P(t)\,\Theta(P(t))\,dt. \tag{17}$$

$$\text{CPS} = C/t \tag{18}$$

The battery equivalent full cycle is calculated using (17) refer to [4], where E, P, and t are battery energy capacity, rated power, and simulation time, respectively. Then the cycle number is converted to CPS (cycle per second) using (18), converted to hours by multiplying with 3600, and to days by multiplying with 5. This value is based on the survey result, Fig. 1. Next, to convert to years, it is multiplied by 365 days. The constraint of this optimization process is the SoC of the battery and SC

$$20\% < \text{SoC}_\text{B} < 100\%$$
$$55\% < \text{SoC}_\text{SC} < 95\% \tag{19}$$

The HESS price (P_{HESS}) is a combination of the SC and DC-DC converter prices, as in Table 5. Whereas the HESS's mass, event is not used as an optimization constraint, it is used to add the mass of the EV.

TABLE 5. HESS components' price and mass.

Components	Price [42]	Euro to USD (Exch. rate 1.09)	Mass [43]
Li-ion battery	420 (€/kWh)	457.34 (USD/kWh)	85.71 (Wh/kg)
DC-DC converter	150 (€/kW)	163.34 (USD/kW)	6.35 (kW/kg)
SC	10 (€/Wh)	10.89 (USD/Wh)	5 (Wh/kg)

IV. RESULTS AND DISCUSSION

The suggested method's effectiveness is confirmed by the simulation test. There are two stages: training and validation. Through training, the optimal sizing and cut-off frequency are determined. The WLTC drive cycle is part of the World-wide Harmonized Light Vehicle Test Procedure (WLTP) for light vehicles. Because WLTP is a worldwide standard for calculating the emissions of traditional, hybrid, and fully electric vehicles in terms of fuel, CO_2, and pollutants [27], this training makes use of it. The speed profile of WLTC and its corresponding load current using the HESS EVs model is shown in Fig. 10. Its performance will be evaluated and contrasted with another approach in various drive cycles during the validation phase.

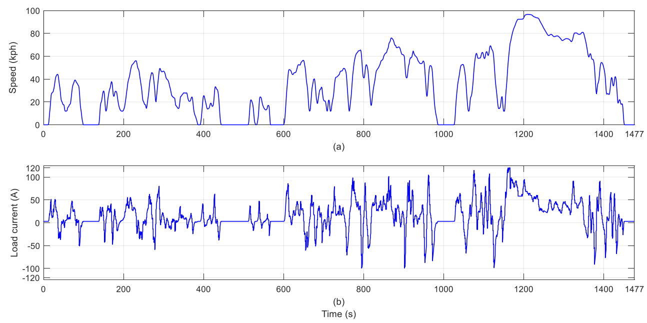


FIGURE 10. WLTC drive cycle: (a) speed (b) load current.

There is no grade or change in height on the track. The starting values for the battery and supercapacitor SoC are 60% and 75%, respectively. This value was selected because it falls between 20–100% and 55–95% for battery and SC SoC limits, respectively. Since the initial SoC may differ in the actual car, an average (typically in the middle) is employed.

A. TRAINING

In this step, the optimal sizing and cut-off frequency of LPF EMS are determined simultaneously using PSO. Based on Fig. 8, the LPF issues are SoC control, finding the cut-off frequency, phase shift effect, and load changes. The SoC control must be added in LPF EMS to make sure the ESS works in the safe SoC limit. The cut-off frequency finding issue is solved using the optimization method, hence the LPF is named Optimal-LPF (OLPF). Next, the PSR is added to reduce the phase shift effect from LPF EMS, this method is called OLPF-PSR. Lastly, load variation is solved using frequency switching between powering and braking based on load current. Since the last method combines all modifications to solve all the mentioned LPF EMS issues, it is named Improved-LPF (ILPF). All OLPF, OLPF-PSR, and ILPF will be compared to show the performance improvement of each approach.

Fig. 11 shows the convergence plot from the optimization process. The PSO process uses 10 particles with 10 iterations, hence a total of 100 simulations are done in each control mode. The use of PSR has a significant effect on reducing the objective function value. The additional frequency switching only has a small effect. For details, results are resumed in Table 6.

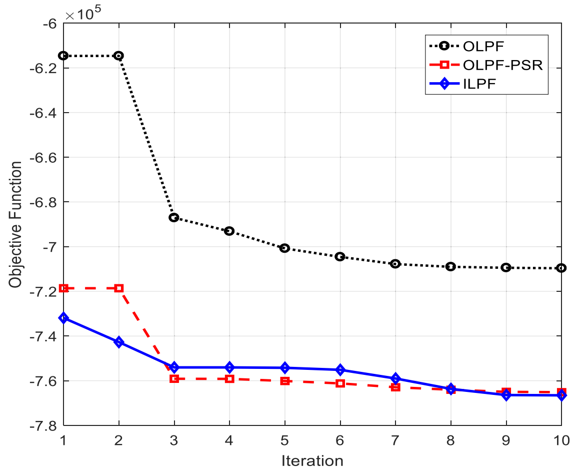


FIGURE 11. Convergence plot.

Two approaches with use PSR (OLPF-PSR and ILPF) give the same HESS size which is 22 Wh and 5000 W for SC and DC-DC converters, respectively. On the other hand, the OLPF gives the same SC size with lower DC-DC converter power. Therefore, the OLPF has the lowest HESS price. Focusing on the objective function value which is the combination of total traveling distance and HESS component price, the proposed ILPF is the best with the lowest objective function value.

TABLE 6. Optimal size and LPF's FC.

Mode	Fc (Hz)	Obj Function	E_{SC} (Wh)	$P_{DC\text{-}DC}$ (W)	P_{HESS} (USD)
OLPF	Fc=0.007	-709,673.30	22	3000	729.60
OLPF-PSR	Fc=0.007	-765,120.65	22	5000	1056.28
ILPF	Fc1=0.010 Fc2=0.007	-766,522.81	22	5000	1056.28

Based on the SC module specification in Table 2, the SC pack requires an 11-series connection to get 176 V and 1 parallel connection to get 22 Wh with a total capacitance of 5.27 F rounded to 5 F. The SC module is used for easy assembly in the real-application event its final specification over the optimal value. Using an SC cell is better to get nearly the same SC pack specification from the optimal SC's energy. However, it may cost more since there are extra costs for assembling the single cell into a pack. In this study, the exact value of SC's energy from the optimization process will be used for the next analysis.

After the optimal size and optimal LPF's cut-off frequency are obtained, a deep analysis of the total traveling distance is done. Table 7 shows the comparison of the optimal size and LPF mode in terms of total traveling distance including autonomy and the estimated battery life. The battery-only was used as the base comparison. Generally, the HESS has lower autonomy and higher estimated battery life compared to the battery-only. The total traveling distance is calculated based on formula (12). It shows that most of the HESS EVs can improve the total traveling distance. The improvement in Table 7 shows that the OLPF makes a negative improvement. This is due to the phase shift effect which causes unnecessary charging and makes the losses higher. After the PSR was added in OLPF-PSR and ILPF, the total traveling distance improvement was achieved. The ILPF gives a bit higher improvement compared to OLPF-PSR. Hence, for the validation step, only the ILPF is used and compared with another EMS method and drive cycles.

TABLE 7. Optimal size and LPF comparison.

Method	Autonomy (km)	Est. Bat. Life (years)	Total traveling dist. (km)	Improve -ment (%)
Battery only	212.334	9.298	720,650	0
OLPF	197.599	9.849	710,380	-1.425
OLPF-PSR	199.589	10.508	765,540	6.229
ILPF	199.660	10.533	767,630	6.519

B. VALIDATION

To verify the performance of the proposed ILPF method, it will be compared with another method. The comparison will be in both trained drive cycles, WLTC, and outside which use UDDS (Urban Dynamometer Driving Schedule), HWFET (Highway Fuel Economy Test Cycle), and a combination of UDDS, HWFET, and WLTC called UHW, depicted in Fig. 12. The distance of the WLTC, UDDS, and HWFET drive cycles are 15.01 km, 12.07 km, and 16.45 km respectively. The ILPF will be compared with the battery-only EV as the base comparison, FLC EMS, and conventional LPF which is the cut-off frequency from the Ragone plot. The LPF has a cut-off frequency of 0.01 Hz.

Table 8 shows the results of performance validation in various drive cycles. In this test, the parameter for comparison is the total traveling distance, while the optimal size is using the optimal size from the training process which is 22 Wh and 5000 W for the SC's size and DC-DC converter size, respectively. The overall results show that the conventional LPF method reduces performance. This is due to high losses in the LPF EMS because of the phase shift effect. While the other EMS for the HESS EV gives the improvement compared to the battery-only system.

TABLE 8. Validation performance comparison.

Methods	WLTC (km)	UDDS (km)	HWFET (km)	UHW (km)
Battery-only	720,650	969,030	272,790	577,440
FLC	720,610	978,990	251,500	562,470
LPF	692,370	951,970	246,930	546,010
ILPF	767,630	1,049,900	251,740	584,620

Starting from WLTC where the tuning or optimization process is done, the proposed ILPF is superior compared to

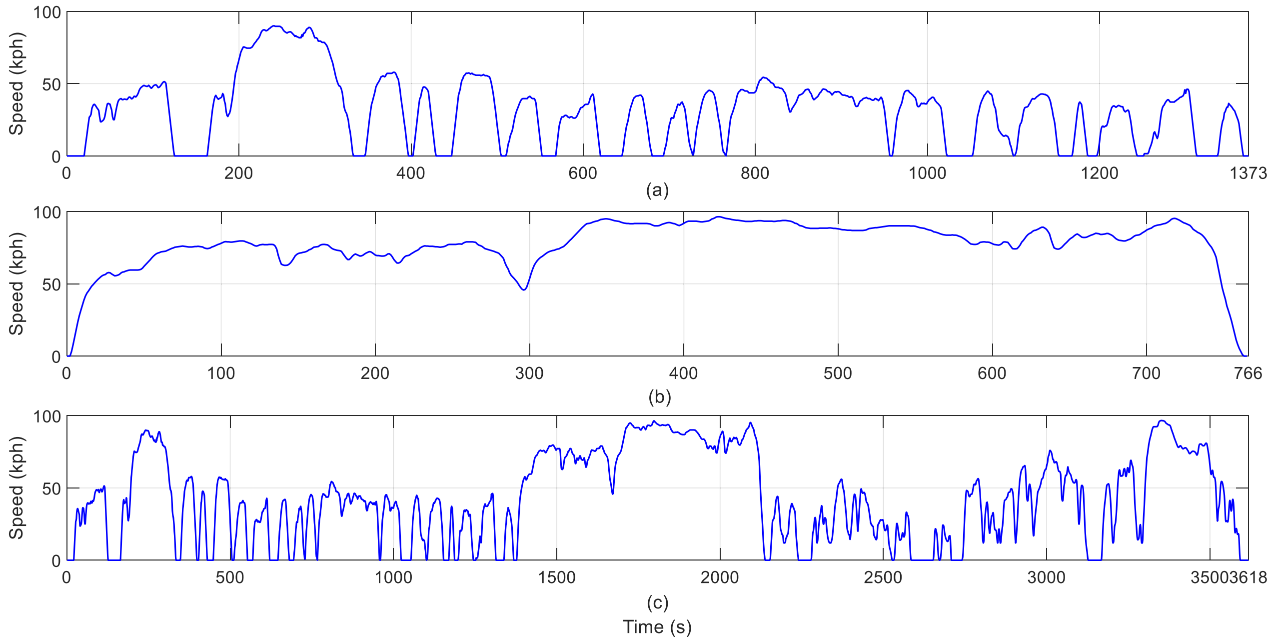


FIGURE 12. Drive cycles: (a) UDDS, (b) HWFET, (c) UHW.

the other methods. WLTC is the drive cycle that combines city, urban, and highway profiles. With the battery-only as the base comparison, the ILPF can improve its performance by 6.519%. On the other hand, the FLC EMS has nearly the same performance as the battery-only EV. Next, in the UDDS drive cycle, which is the city profile, the proposed ILPF is still superior with an improvement of up to 8.345%. At this condition, the FLC EMS gives an improvement of only 1.028%.

However, in the highway scenario represented by the HWFET drive cycle, all EMS methods yield lower performance than the battery-only configuration. This reduction occurs because highway driving typically involves steady-state conditions with minimal acceleration and deceleration events, leading to the supercapacitor's role as an energy buffer being underutilized. As a result, the supercapacitor does not operate optimally, diminishing its effectiveness in supporting energy demands. This limitation suggests the need for potential solutions, such as drive cycle prediction hence the SC power can be allocated properly for city, urban, or highway.

Finally, in the UHW drive cycle, the proposed ILPF still gives the performance improvement by up to 1.243%. On the other side, the FLC EMS decreases the performance by 2.592%. From these results, it can be concluded that HESS provides significant advantages in city and urban profiles, where acceleration and deceleration frequently occur, but faces challenges on highways.

The detailed analysis of the energy sharing between the battery and SC will be discussed for the WLTC drive cycle as shown in Fig 13. Positive current means that the ESS supplies the EV. On the other hand, negative current means braking, and power flows back because of regenerative braking. In the battery-only, all the EV power is supplied by the battery; hence the SC current is zero. In the battery current, the FLC EMS has the same pattern as the battery-only EV since the working principle of this method is scaling. As a result, the current of the battery is reduced and sent to the SC, the reduction is based on the scaling gain as the output from FLC EMS.

On the other hand, the battery power from the LPF-based method is different from the battery-only EV since the high frequency is sent to the SC. Most of the HESS EV's battery current is positive. This is the result of SC acting as the hub for energy regeneration. The SC serves as an energy buffer for batteries that can handle the high frequency of current demand, hence, its current profile is more varied.

In the LPF EMS, since the high frequency is sent to the SC, the battery current looks smooth. Conversely, in the FLC EMS, battery power references maintain a high frequency while decreasing the peak power. Consequently, the LPF EMS uses the SC power more often. The charge and discharge of the battery and SC occur more frequently with the LPF method than with the FLC EMS.

When comparing the LPF and ILPF, the close cut-off frequencies mean that both EMS techniques show similar battery and supercapacitor current patterns. However, the proposed ILPF, with its phase-shift reduction mechanism, helps to minimize unnecessary power exchange between the battery and supercapacitor. This effect is illustrated in Fig. 14. In the LPF when the load current becomes negative, the battery current remains positive and charges the supercapacitor.

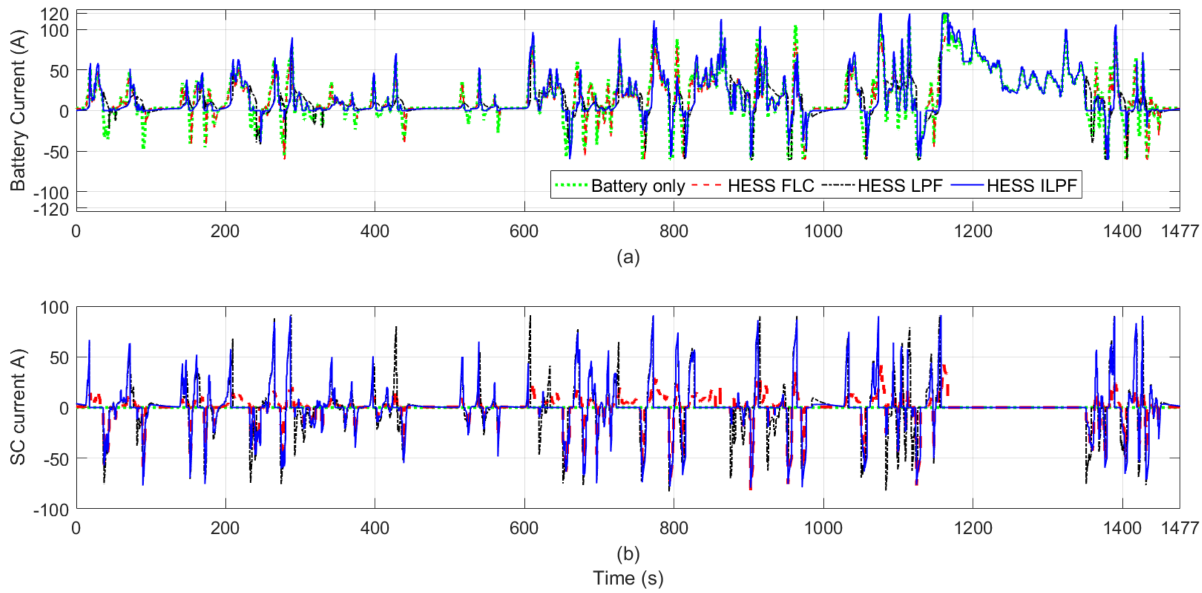


FIGURE 13. Current (a) Battery (b) SC.

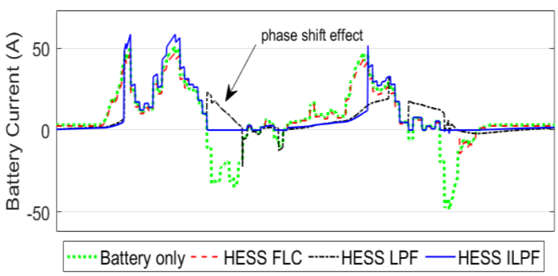


FIGURE 14. Phase shift effect of LPF EMS.

In contrast, the ILPF allows the supercapacitor to charge exclusively from regenerative power, preventing unnecessary power exchange that can increase losses due to the DC-DC converter on the supercapacitor side. The FLC EMS also drops to zero, as it only activates for positive current, sending all negative current to the supercapacitor.

V. CONCLUSION

The optimal sizing and energy management of HESS EVs have been proposed and validated through simulation. The LPF EMS was chosen for its simplicity and practicality. By employing the PSO method, the optimal sizing for the supercapacitor and its associated DC-DC converter has been determined, leading to the development of the ILPF that addresses key issues associated with traditional LPF EMS.

The objective function focuses on balancing the total traveling distance with the component costs of the HESS. In the verification process, the proposed ILPF was compared against both FLC and conventional LPF methods across various drive cycles. The simulation results demonstrate that the ILPF significantly outperforms both the conventional LPF and FLC EMS, achieving up to an 8.345% improvement in city driving cycles compared to battery-only EVs. However, the performance of HESS systems declines on highway routes, indicating a preference for HESS EVs in the city and urban environments.

This study acknowledges several limitations, including the lack of consideration for calendar aging and battery cooling, as well as the omission of installation and maintenance costs associated with HESS. While the ILPF effectively reduces the phase shift effect, the inherent delay characteristic of the LPF remains unavoidable. Future research should aim to address these limitations to provide more comprehensive results and insights into the practical implementation of HESS in electric vehicles.

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